ECG Arrhythmia

Classifying Heartbeats, and the Inter-Patient Generalization Gap

Capstone Project Report

A 1-D neural network that sorts heartbeats into four clinical categories — and an honest look at why 96% in the lab becomes 44% on unseen patients.

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1. Executive Summary

This project classifies individual heartbeats from an ECG into four AAMI categories — Normal, Supraventricular, Ventricular, and Fusion — using a compact **1-D convolutional network** (77,000 parameters). More than the model, the project is a case study in honest evaluation.

During training, validation accuracy looked excellent. But on a **patient-independent** test set the honest result is **70% accuracy and 44% macro-recall**: the model catches 86% of ventricular beats but only 9% of supraventricular ones. That gap is the lesson.

2. Introduction & Motivation

Arrhythmia detection from ECG is a classic biosignal problem. The aim was to build the full pipeline — data, model, on-device export, browser demo — while treating the evaluation with the rigour the domain demands.

3. Data

The MIT-BIH Arrhythmia Database (PhysioNet) provides 48 half-hour, single-lead recordings with a cardiologist's label on every beat (~100,000 beats). Each beat was cut into a 260-sample window centred on its peak and normalised. Beats map to four AAMI classes; the tiny paced class was excluded.

> Key decision: the standard **inter-patient split** (de Chazal) was used — training and test recordings come from **different patients**, so results reflect generalisation to new people, not memorised individuals.

4. Methodology

A four-block 1-D CNN (77k parameters) was trained with an inverse-frequency weighted loss, because ~90% of beats are normal and a naive model would ignore the rare, important classes. The checkpoint was selected by macro-recall, not accuracy.

5. Results

On the untouched test patients:

Class	Recall	Note
Normal	72%	Majority class
Supraventricular	9%	Nearly missed entirely
Ventricular	86%	Caught well
Fusion	9%	Very rare, hard

Overall: 70% accuracy, **44% macro-recall**, ROC-AUC 0.958 at the window level.

Why S and F fail

A ventricular beat has a visibly mis-shapen waveform, so a shape-reading model spots it. A supraventricular beat looks almost normal — what marks it is its **timing** (it arrives early). The model sees each beat in isolation, so it is blind to the one feature that defines the class it fails on.

6. Deployment

The model was exported to a self-contained ONNX file (325 KB) that runs in the browser via WebAssembly, and to TensorFlow Lite for an Android app ('ECG Check'), both verified to match the PyTorch model to within 1e-5. A visitor picks a real unseen-patient beat and watches it classified on-device.

7. Limitations & Ethical Considerations

- Not a medical device — an educational study on a research dataset.
- Trained on curated research beats; real-world signals are noisier.
- The honest 44% macro-recall shows the model is not deployable as-is — which is the point.

8. Future Work

- Add RR-interval (beat-timing) features — the missing signal for supraventricular beats.
- Report a full confusion analysis and calibrated confidence.